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Quadrotor Dual Quaternion Control

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Abstract— **This paper introduces the design and practical validation of a quaternion control scheme to globally stabilize a quadrotor aerial vehicle. First an attitude control law is proposed to stabilize the vehicle's heading, then a position control law is developed to stabilize the vehicle around a desired point. Using position references, a smooth trajectory is calculated for the attitude controller such that the position of the vehicle becomes stable. The proposed control algorithm allows a linear quadrotor system's behavior. The closed-loop system is then numerically simulated to corroborate stability.**

I. INTRODUCTION

A quadrotor is a popular platform in UAV (Unmanned Aerial Vehicles) research because of the relatively simple dynamic model and its now low-cost implementation. Although the dynamic model could be considered simple, the non-linearities introduced by using Euler's angles representation makes the analysis of the system and the design of control laws cumbersome. The Euler's angles approach is preferred because of its intuitiveness, in [3], the authors describe the design, modeling, control, and simulation of a quadrotor by using this approach but it is apparent that it makes the platform's model loose its mathematical simplicity. A more computationally stable and more simple mathematical tool for modeling these dynamics are quaternions.

Quaternions are a type of hyper-complex numbers that can be used to describe rotations in 3D space. Its computational stability comes from the fact that only unit quaternions are required to describe rotations, reducing numeric errors when they are normalized. They lack the gimball-lock effect inherent to Euler's angles because of its relationship to the axis-angle representation. Their mathematical simpleness comes from the fact that no trigonometric functions are required directly when they are used to model a rigid body's dynamics, only arithmetic functions like quaternion products (which are computationally efficient).

In [5] the authors build a quaternion-based model for an hexa-rotor vehicle, which can be extended to build a similar model for a quad-rotor UAV. In [6], the authors build a quaternion-based model and control for a robot manipulator. In [6], [7], [8], and [9] , the authors establish the mathematical tools for quaternion algebra and quaternion kinematics. In [2], the authors proposed a linear control using dual quaternions, however their example can only be applied to fully actuated systems, which is not the case in multi-copter vehicles.

The main contribution of this article is the design of a quadcopter control law that uses dual quaternions in an under-actuated system, which facilitate the kinematic and dynamic analysis of the coupled translational and rotational dynamics on the quadrotor platform. The effects of a state feedback are analyzed on a dual quaternion model. A trajectory relying on the position and linear velocity can be obtained that guarantees stability and then it can be fed to the attitude reference, resulting in a PD control law that fully stabilizes the system.

The paper is organized as follows: a dual quaternions background and the dynamical model using this approach are given in section II, the control algorithms are introduced in section III, the main graphs representing the behavior of the closed-loop system are depicted in section IV, the experimental validation of the algorithm and the prototype used for the tests is described in section V, and finally some discussions about this work are given in section VI.

II. DUAL QUATERNION MODELING

A. Background

1) Quaternions: As explained in [1], quaternions are hyper-complex numbers that belong to the space $\mathbb{H}$. They can be represented as a sum of a scalar component along with an imaginary vector. See [6], [7], [9].

$$\boldsymbol{q} \in \mathbb{H}; \bar{q} \in \mathbb{R}^3; q_0 \in \mathbb{R}$$
$$\boldsymbol{q} = q_0 + \bar{q} = q_0 + \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{1}$$

where $\boldsymbol{q}$ is a given quaternion, $\bar{q}$ is the complex vectorial part of $\boldsymbol{q}$, and q_0 is the scalar part of $\boldsymbol{q}$.

Let $\boldsymbol{r} \in \mathbb{H}$; $\bar{r} \in \mathbb{R}^3$; $r_0 \in \mathbb{R}$. Thus, the quaternion operations are defined as: (see [1])

- Product: $\boldsymbol{q} \otimes \boldsymbol{r} = (q_0\, r_0 - \bar{q} \cdot \bar{r}) + (r_0\, \bar{q} + q_0\, \bar{r} + \bar{q} \times \bar{r})$
- Conjugate: $\boldsymbol{q}^* = q_0 - \bar{q}$
- Norm: $||\boldsymbol{q}|| = \sqrt{\boldsymbol{q} \otimes \boldsymbol{q}^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}$

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- Inverse: $\boldsymbol{q}^{-1} = \dfrac{\boldsymbol{q}^*}{||\boldsymbol{q}||}$
- Natural logarithm mapping:

$$\ln \boldsymbol{q} = \begin{cases} \ln ||\boldsymbol{q}|| + \dfrac{\overline{q}}{||\overline{q}||} \arccos \dfrac{q_0}{||\boldsymbol{q}||}, & ||\overline{q}|| \neq 0 \\ \ln ||\boldsymbol{q}||, & ||\overline{q}|| = 0 \end{cases}$$

In this work, only unit quaternions are used for the attitude representation, thus, $\boldsymbol{q}^{-1} = \boldsymbol{q}^*$. Then the logarithmic mapping can be reduced to:

$$\ln \boldsymbol{q} = \begin{cases} \dfrac{\overline{q}}{||\overline{q}||} \arccos q_0, & ||\overline{q}|| \neq 0 \\ 0, & ||\overline{q}|| = 0 \end{cases}$$

This mapping is useful to change any unit quaternion to the axis angle representation, in which the quaternion is represented as an axis over which the vehicle will rotate, and a rotation angle.

$$\overline{\theta} = 2 \ln \boldsymbol{q}, \quad \dot{\overline{\theta}} = \omega$$

where $\bar{\theta} \in \mathbb{R}^3$ is the vector which contains the axis-angle representation, in which $\frac{\bar{\theta}}{||\bar{\theta}||}$ describes the unitary axis over which the rotation is applied, $||\bar{\theta}||$ represents the magnitude of the rotation, and ω is the vehicle's angular velocity.

- Quaternion Rotation:
The rotation of a 3D vector from one reference frame to another can be accomplished with the three quaternion product

$$\begin{aligned} &||\boldsymbol{q}|| = 1; \; v, v' \in \mathbb{R}^3 \\ &v' = L(\boldsymbol{q}, v) = \boldsymbol{q}^* \otimes v \otimes \boldsymbol{q} \end{aligned} \tag{2}$$

The rotation operation $L(\boldsymbol{q}, v)$ is a linear function as it preserves the principles of superposition and homogeneity present in the unit quaternion algebra product.

2) Dual Numbers: Dual numbers are defined in [2] as:

$$a, b \in \mathbb{R}, \quad \hat{a} = a + \epsilon b, \quad \epsilon \neq 0, \quad \epsilon^2 = 0$$

where ϵ is a nullpotent term, and a and b are called *real part* and *dual part* respectively.

Define three-dimensional dual vectors, which are a generalization of dual numbers:

$$\bar{v}_r, \bar{v}_d, \bar{k}_r, \bar{k}_d \in \mathbb{R}^3, \quad \hat{\bar{v}} = \bar{v}_r + \bar{v}_d \epsilon, \quad \hat{\bar{k}} = \bar{k}_r + \bar{k}_d \epsilon$$

Then the dot-product for dual vectors is defined as:

$$\hat{\bar{k}} \cdot \hat{\bar{v}} = K_r \bar{v}_r + K_d \bar{v}_d, \tag{3}$$

where $K_r, K_d \in \mathbb{R}^{3\times3}$ are diagonal matrices with entries k_{r1}, k_{r2}, k_{r3} and k_{d1}, k_{d2}, k_{d3} respectively.

3) Dual Quaternions: When the components of a dual number belong to $\mathbb{H}$, they are called dual quaternions, which are expressed as:

$$\hat{q} = \boldsymbol{q}_r + \boldsymbol{q}_d \epsilon$$

Dual quaternions have the following main operations:

- Sum: $\hat{q}_1 + \hat{q}_2 = \boldsymbol{q}_{1r} + \boldsymbol{q}_{2r} + [\boldsymbol{q}_{1d} + \boldsymbol{q}_{2d}]\epsilon$
- Product:

$$\hat{q}_1 \otimes \hat{q}_2 = \boldsymbol{q}_{1r} \otimes \boldsymbol{q}_{2r} + [\boldsymbol{q}_{1r} \otimes \boldsymbol{q}_{2d} + \boldsymbol{q}_{1d} \otimes \boldsymbol{q}_{2r}]\epsilon$$

- Conjugate: $\hat{q}^* = \boldsymbol{q}_r^* + \boldsymbol{q}_d^* \epsilon$
- Norm: $||\hat{q}||^2 = \hat{q} \otimes \hat{q}^*$

When $||\hat{q}||^2 = 1 + 0\epsilon$, then the dual quaternion is called *unit dual quaternion*

- Inverse: $\hat{q}^{-1} = (1/||\hat{q}||^2) \otimes \hat{q}^*$

For unit dual quaternions, the inverse is reduced to:

$$\hat{q}^{-1} = \hat{q}^*$$

- Dual quaternion transformation:

Unit dual quaternions can be used to represent pose transformations (rotation and translation simultaneously). Let $\boldsymbol{q}$ be the rotation of a rigid body and T be its translation with respect of its body reference. We suppose that, mathematically, the rotation happens first, and then the translation occurs.

$$\hat{q} = \boldsymbol{q} + \frac{1}{2} \boldsymbol{q} \otimes T_B \epsilon \tag{4}$$

The complete pose transformation of a dual vector $\hat{\bar{x}} = \bar{d} + (\bar{d} \times \bar{p})\epsilon$ where $\bar{d} \in \mathbb{R}^3$ represents the direction of a line and $\bar{p} \in \mathbb{R}^3$ is its position (in the body reference frame), into $\hat{\bar{x}}' \in \mathbb{R}^3$, which usually represents the vector in the body frame, can be represented using dual quaternions as:

$$\hat{\bar{x}}' \epsilon = \hat{q}^* \otimes \hat{\bar{x}} \otimes \hat{q} \tag{5}$$

- Natural logarithm mapping:

$$\ln \hat{q} = \frac{1}{2}(\bar{\theta} + T_B \epsilon)$$

where $2 \ln \boldsymbol{q} = \bar{\theta} \in \mathbb{R}^3$ is the vector which contains the axis-angle representation, and T is the translation.

B. Dual Quaternion Kinematics

The kinematics of a rigid body in terms of its orientation, position and twist is given in [2] by

$$\dot{\hat{q}} = \frac{1}{2}\hat{q} \otimes \hat{\xi} \tag{6}$$

where $\hat{\xi}$ is the twist (combination of rotational and translational velocities), which is a dual quaternion given by:

$$\hat{\xi} = \omega + [\omega \times T_B + \dot{T}_B]\epsilon \tag{7}$$

where $\omega = [\omega_x, \omega_y, \omega_z]^T$ is the angular velocity vector in the body frame. T_B is the 3D vector describing the rigid body's position in the body frame. $\dot{T}_B$ is another vector describing the object's translational velocity with respect to the body frame.

Note that any vector in $\mathbb{R}^3$ can be expressed as a quaternion with its real part equal to zero.

C. Quadrotor Dual Quaternion Dynamic Model

The dynamic model of a quadcopter using the Newton-Euler approach could be described by its rotational and translational dynamics.

The rotational dynamics is given by

$$J\dot{\omega} + \omega \times (J\omega) = \tau \tag{8}$$

where $J \in \mathbb{R}^{3\times3}$ is the inertia matrix of the body, and $\tau \in \mathbb{R}^3$ is the vector that contains the torques applied to the body, these torques are the sum of the control input torques and the external torques, then we can write that $\tau = \tau_u + \tau_{ext} = \left[\tau_{u_x}\, \tau_{u_y}\, \tau_{u_z}\right]^T + \tau_{ext}$.

Similarly, the translational dynamic relative to its body frame is

$$m\ddot{T}_B = F \tag{9}$$

where $m \in \mathbb{R}$ is the mass, and $F \in \mathbb{R}^3$ is the vector that contains the sum of the control input forces and the external forces in the body frame, thus $F = F_u + F_{ext}$.

Differentiating equation (7):

$$\dot{\hat{\xi}} = \dot{\omega} + [\dot{\omega} \times T_B + \omega \times \dot{T}_B + \ddot{T}_B]\epsilon \tag{10}$$

Substituting equations (8) and (9) in equation (10), the following dynamic model is obtained:

$$x = \begin{bmatrix} \hat{q} \\ \hat{\xi} \end{bmatrix}, \quad \dot{x} = \begin{bmatrix} \dot{\hat{q}} \\ \dot{\hat{\xi}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\hat{q} \otimes \hat{\xi} \\ \hat{F} + \hat{u} \end{bmatrix} \tag{11}$$

where

$$\left\{ \begin{array}{rcl} \hat{F} &=& a + (a \times T_B + \omega \times \dot{T}_B)\epsilon \\ \hat{u} &=& J^{-1}\tau + [J^{-1}\tau \times T_B + m^{-1}F]\epsilon \\ a &=& -J^{-1}(\omega \times J\omega) \end{array} \right. \tag{12}$$

The relationships between the input torques and forces are

$$\begin{bmatrix} F_u \\ \tau_{u_x} \\ \tau_{u_y} \\ \tau_{u_z} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{4} k_i\,\omega_i^2 \\ l\left(k_1\,\omega_1^2 - k_2\,\omega_2^2 - k_3\,\omega_3^2 + k_4\,\omega_4^2\right) \\ l\left(k_1\,\omega_1^2 + k_2\,\omega_2^2 - k_3\,\omega_3^2 - k_4\,\omega_4^2\right) \\ \sum_{i=1}^{4} \tau_i\,(-1)^i \end{bmatrix} \tag{13}$$

where $k_i\,\omega_i^2$ defines the thrust of the propeller of motor i with respect to its angular velocity ω_i. For example, ω_1 is the angular velocity of the first motor. l is the distance from the center of mass to the motor axis of action and τ_i denotes the torque of motor i.

III. Control Algorithm

A. Proposed Control Law

A control is proposed such that the non-linearities of subsystem $\dot{\hat{\xi}} = \hat{F} + \hat{u}$ can be canceled:

$$\hat{u} = 2\hat{k}_p \cdot \ln\hat{q} + \hat{k}_v \cdot \hat{\xi} - \hat{F} \tag{14}$$

with $\hat{k}_p = \bar{k}_{pr} + \bar{k}_{pd}\epsilon = [\bar{k}_{pr1}, \bar{k}_{pr2}, \bar{k}_{pr3}]^T + [\bar{k}_{pd1}, \bar{k}_{pd2}, \bar{k}_{pd3}]^T\epsilon$ y $\hat{k}_v = \bar{k}_{vr} + \bar{k}_{vd}\epsilon = [\bar{k}_{vr1}, \bar{k}_{vr2}, \bar{k}_{vr3}]^T + [\bar{k}_{pv1}, \bar{k}_{pv2}, \bar{k}_{pv3}]^T\epsilon$

Theorem 1: The control in equation (14) assures that the dual quaternion $\hat{q}$ (with a unique representation by assuring that $q_0 \geq 0$) in (11) converges asymptotically to the identity dual quaternion $\hat{q}_0 = 1 + [0,0,0]^T + \left[0 + [0,0,0]^T\right]\epsilon$, using appropriate $\hat{k}_p$ and $\hat{k}_v$.

Proof: Introducing (14) into (11):

$$\begin{bmatrix} \dot{\hat{q}} \\ \dot{\hat{\xi}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\hat{q} \otimes \hat{\xi} \\ 2\hat{k}_p \cdot \ln\hat{q} + \hat{k}_v \cdot \xi \end{bmatrix}$$

Taking the subsystem:

$$\dot{\hat{\xi}} = 2\hat{k}_p \cdot \ln\hat{q} + \hat{k}_v \cdot \hat{\xi} \tag{15}$$

Applying the definition of the logarithmic mapping to (15), we obtain:

$$\dot{\hat{\xi}} = 2\hat{k}_p \cdot \frac{1}{2}(\bar{\theta} + T_B\epsilon) + \hat{k}_v \cdot \hat{\xi}$$

$$\dot{\hat{\xi}} = \hat{k}_p \cdot (\bar{\theta} + T_B\epsilon) + \hat{k}_v \cdot \hat{\xi}$$

$$\dot{\hat{\xi}} = \hat{k}_p \cdot (\bar{\theta} + T_B\epsilon) + \hat{k}_v \cdot (\omega + [\omega \times T_B + \dot{T}_B]\epsilon) \tag{16}$$

Using (16) and (10), it can be obtained (using the dual vector dot-product) that the rotational acceleration is given by:

$$\dot{\omega} = K_{pr}\bar{\theta} + K_{vr}\omega \tag{17}$$

And the translational acceleration by:

$$\ddot{T}_B = K_{pd}\,T_B + K_{vd}(\omega \times T_B + \dot{T}_B) - (\dot{\omega} \times T_B + \omega \times \dot{T}_B) \tag{18}$$

Taking into account that $\omega = \dot{\bar{\theta}}$, then we obtain the following equation for the rotational acceleration.

$$\ddot{\bar{\theta}} - K_{vr}\dot{\bar{\theta}} - K_{pr}\bar{\theta} = 0 \tag{19}$$

Using Laplace transformation, we can obtain the following expression:

$$\begin{aligned} \bar{\Theta}s^2 - K_{vr}\bar{\Theta}s - K_{pr}\bar{\Theta} = 0 \\ s^2 - K_{vr}s - K_{pr} = 0 \end{aligned} \tag{20}$$

If $\bar{k}_{vr}$ and $\bar{k}_{pr}$ are selected such that the roots of (20) are located in the real negative semi-plane, then $\bar{\Theta}$ as well as ω, and $\dot{\omega}$ will converge asymptotically to zero.

For the translational acceleration, the following procedure is used:

We can gather all the forces that are originated from the rotational velocity and acceleration defining

$$F(\omega, \dot{\omega}) = -K_{vd}(\omega \times T_B) + \dot{\omega} \times T_B + \omega \times \dot{T}_B \tag{21}$$

It was already proven that $\bar{\theta}$, ω, and $\dot{\omega}$ converge asymptotically to zero, so it can be said that when $t \to \infty$, then $F(\omega, \dot{\omega}) \to 0$. Then there exists a finite time such that equation (21) is small and can be neglected. This would imply that equation (18) could be reduced to:

$$\ddot{T}_B = K_{pd}T_B + K_{vd}\dot{T}_B \tag{22}$$

Using Laplace transform, we obtain the following polynomial:

$$\begin{aligned} \mathbf{T_B}s^2 - K_{vd}\mathbf{T_B}s - K_{pd}\mathbf{T_B} = 0 \\ s^2 - K_{vd}s - K_{pd} = 0 \end{aligned} \tag{23}$$

Thus, if $\bar{k}_{vd}$ and $\bar{k}_{pd}$ are selected such that the roots of (23) are located in the real negative semi-plane, then T_v as well as $\dot{T}_v$, and $\ddot{T}_v$ converge asymptotically to zero. ■

B. Application in a Quadrotor Platform

It is well known that quadrotors are sub-actuated systems. Although the input torques can act over the three axes (x, y, z), the input forces can only act in one body axis. We can see this in (13).

In order to have a similar stable behavior like in a fully-actuated system, a trajectory for the attitude must be calculated using equation (14), and a relationship between a desired force in the inertial frame and a reference in orientation.

Introducing (7) into (14), it follows that:

$$\begin{aligned} \hat{u} = {} & 2\hat{k}_p \cdot \ln \hat{q} + \hat{k}_v \cdot (\omega + [\omega \times T + \dot{T}]\epsilon) - \hat{F} \\ = {} & J^{-1}\tau + [J^{-1}\tau \times T + m^{-1}F]\epsilon \end{aligned}$$

Then

$$\begin{aligned} &\hat{k}_p \cdot (\bar{\theta} + T\epsilon) + \hat{k}_v \cdot (\omega + [\omega \times T + \dot{T}]\epsilon) \\ &-a - (a \times T + \omega \times \dot{T})\epsilon \\ &= J^{-1}\tau + [J_1^{-1}\tau \times T + m^{-1}F1]\epsilon \end{aligned}$$

$$\begin{aligned} &K_{pr}\bar{\theta} + K_{pd}T_v\epsilon + K_{vr}\omega + K_{vd}[\omega \times T + \dot{T}_v]\epsilon \\ &-a - (a \times T + \omega \times \dot{T})\epsilon \\ &= J^{-1}\tau + [J^{-1}\tau \times T + m^{-1}F]\epsilon \end{aligned}$$

Separating the real and dual parts from (III-B) we obtain the expressions for the input torques and forces:

$$\begin{aligned} \tau = \tau_u + \tau_{ext} = J(K_{pr}\bar{\theta} + K_{vr}\omega - a) \\ \tau_u = J(K_{pr}\bar{\theta} + K_{vr}\omega - a) - \tau_{ext} \end{aligned} \tag{24}$$

$$\begin{aligned} F = F_u + F_{ext} = m\Big(&K_{pd}T_v + K_{vd}[\omega \times T_v + \dot{T}_v] \\ &-(a_1 \times T_v + \omega \times \dot{T}_v) - J_1^{-1}\tau_1 \times T_v\Big) \end{aligned}$$

$$\begin{aligned} F_u = m\Big(&K_{pd}T_v + K_{vd}[\omega \times T_v + \dot{T}_v] \\ &-(a_1 \times T_v + \omega \times \dot{T}_v) - J_1^{-1}\tau_1 \times T_v\Big) - F_{ext} \end{aligned} \tag{25}$$

Equation (24) determines the torques that will be applied. As it was shown, there is a finite time when the angular velocity and acceleration can be deprecated because when $t \to \infty$, then $\omega \to 0$, $\dot{\omega} \to 0$, and $\tau_1 \to 0$ y $a \to 0$. Therefore the equation (25) can be reduced to:

$$\begin{aligned} F_u = {} & m\Big(K_{pd}T_v + K_{vd}[\omega \times T_v + \dot{T}_v] \\ & -(a_1 \times T_v + \omega \times \dot{T}_v)\Big) - F_{ext} \end{aligned} \tag{26}$$

For our quadrotor platform, it can be considered that F_{ext} is the gravity force in the body reference given by $F_{ext} = m\boldsymbol{q}_v^* \otimes \bar{g} \otimes \boldsymbol{q}_v$

$$\begin{aligned} F_u = {} & m\Big(K_{pd}T_v + K_{vd}[\omega \times T_v + \dot{T}_v] \\ & -(a_1 \times T_v + \omega \times \dot{T}_v)\Big) - m\boldsymbol{q}_v^* \otimes \bar{g} \otimes \boldsymbol{q}_v \end{aligned} \tag{27}$$

It is important to note that the desired control force that is necessary to move the body to the desired reference is given in equation (27) and is referenced to the body frame. This force can be expressed in the inertial frame by defining:

$$F_{uI} \triangleq \boldsymbol{q} \otimes F_u \otimes \boldsymbol{q}^* \tag{28}$$

Remembering that the quadrotor is a under-actuated system and it is considered that the main frame acts only on one main axis of the body frame, the actual force exerted by the motors is given by:

$$\bar{F}_{th} = \begin{bmatrix} 0 \\ 0 \\ F_{th} \end{bmatrix} = bF_{th} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} F_{th} \tag{29}$$

A trajectory taken from [1] is now defined such that the force exerted by the vehicle from (29) coincides in the inertial frame with the required force from (28):

$$\begin{aligned} \boldsymbol{q}_t' &= m^{-1}\left(b \cdot F_{uI} + ||F_{uI}||\right) + m^{-1} b \times F_{uv} \\ \boldsymbol{q}_t &= \frac{\boldsymbol{q}_t'}{||\boldsymbol{q}_t'||} \\ \bar{F}_{th} &= b\,||F_{uI}|| \end{aligned} \tag{30}$$

where F_{uv} is the current thrust force vector with respect to the inertial frame.

IV. Simulations results

The dual quaternion operations were previously defined using NumPy's vector and cross products, and then included in a custom library. The vehicle's parameters, the control gains and the initial values $\hat{q}_{ini}$ were considered as:

$$\begin{aligned} J &= \begin{bmatrix} 0.0118976 & 0 & 0 \\ 0 & 0.0118976 & 0 \\ 0 & 0 & 0.0218816 \end{bmatrix} \\ \hat{q}_{ini} &= \boldsymbol{q}_{ini} + \frac{\boldsymbol{q}_{ini} \otimes T_{ini}}{2}\epsilon \\ \boldsymbol{q}_{ini} &= \frac{1 + \begin{bmatrix} 0 & 0 & \frac{\pi}{10} \end{bmatrix}^T}{\text{norm}\left(1 + \begin{bmatrix} 0 & 0 & \frac{\pi}{10} \end{bmatrix}^T\right)} \\ T_{ini} &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T \\ \dot{T}_{ini} &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\ \omega_{ini} &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\ T_d &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\ \dot{T}_d &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\ m &= 1.3\,kg \\ \boldsymbol{q}_e &:= \boldsymbol{q} \otimes \boldsymbol{q}_d^* \\ \hat{k}_p &= \begin{bmatrix} 100 \\ 100 \\ 3.16 \end{bmatrix} + \begin{bmatrix} 3.1622 \\ 3.1622 \\ 3.1622 \end{bmatrix}\epsilon \\ \hat{k}_v &= \begin{bmatrix} 14.142 \\ 14.142 \\ 2.51 \end{bmatrix} + \begin{bmatrix} 2.7063 \\ 2.7063 \\ 2.7063 \end{bmatrix}\epsilon \end{aligned} \tag{31}$$

1) Dual Quaternion Stabilization: The simulation results were divided in order to better understand them in a more intuitive manner. The attitude is shown in quaternion form and the position as a vector in the body frame to better understand the behavior of the quadrotor.

An initial position T_{ini} was given, then a desired equilibrium point T_d was proposed as a reference to globally stabilize the system. Figures 1 and 2 show that the system's position exponentially converges from the initial condition to the reference.

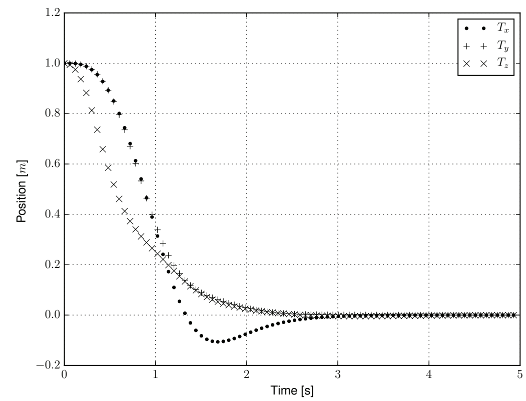

Fig. 1. Quadcopter translational position (p).

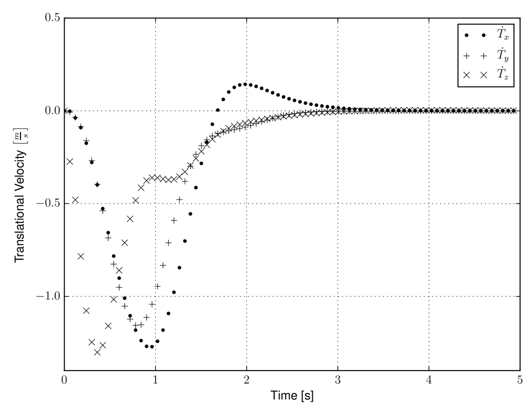

Fig. 2. Quadcopter translational velocity $(\dot{p})$.

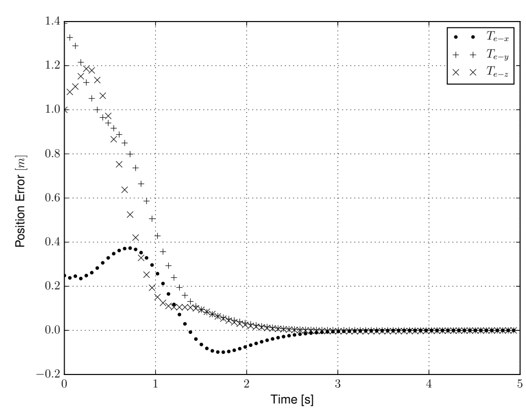

Fig. 3. Quadcopter translational position error.

The resulting error is used to compute the attitude reference trajectory $\boldsymbol{q}_d$ shown as $\boldsymbol{q}_t$ in section III-B, see Figure 4.

Figure 5 shows the system inputs τ_{ux}, τ_{uy}, and τ_{uz}. It can be noted that they converge to zero when the system stabilizes in the desired trajectory.

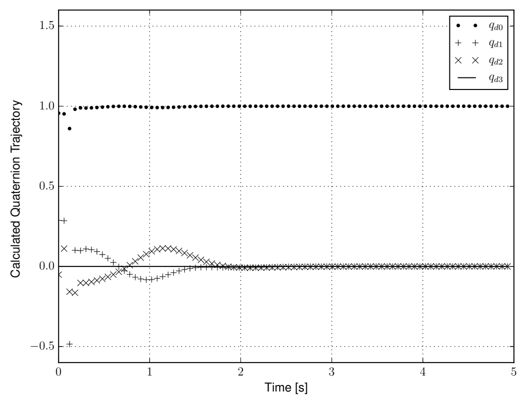

Fig. 4. Quadcopter attitude reference (q_d).

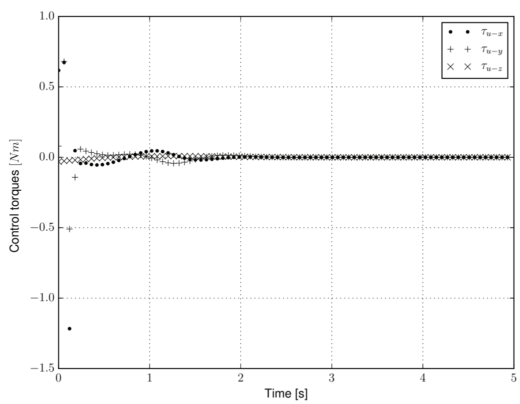

Fig. 5. Quadcopter input.

It can also be seen in Figure 6 that the quaternion stabilizes at the point $\boldsymbol{q} = 1$. Similarly, Figure 7 shows that the angular velocities converge to zero. This means that the vehicle follows the trajectory $\boldsymbol{q}_d$ complimented with the desired reference around the "z" axis, thus stabilizing the dual quaternion.

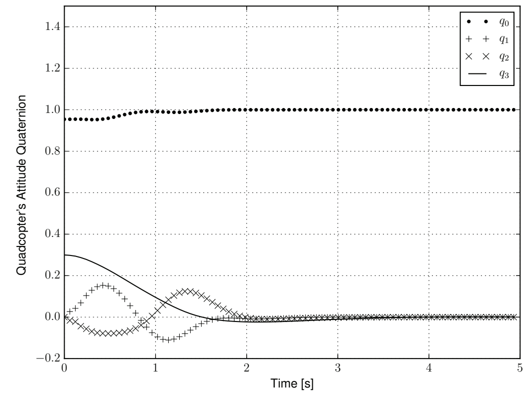

Fig. 6. Quadcopter attitude (q).

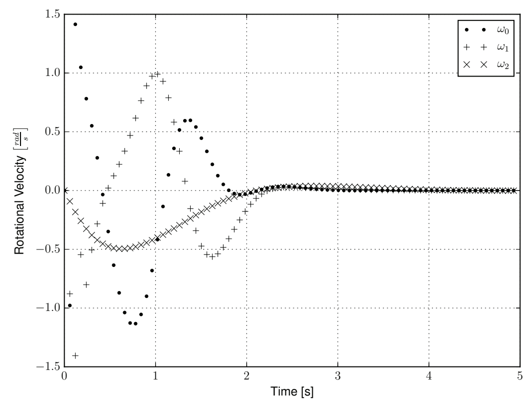

Fig. 7. Quadcopter rotational velocity ω.

Figures 8 and 9 show the attitude and angular velocity errors respectively. It can be seen that $\boldsymbol{q}_e$ converges to the unit quaternion while ω_e stabilizes in zero.

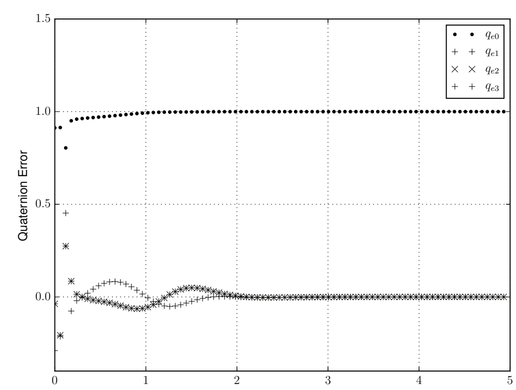

Fig. 8. Quadcopter attitude error (q_e).

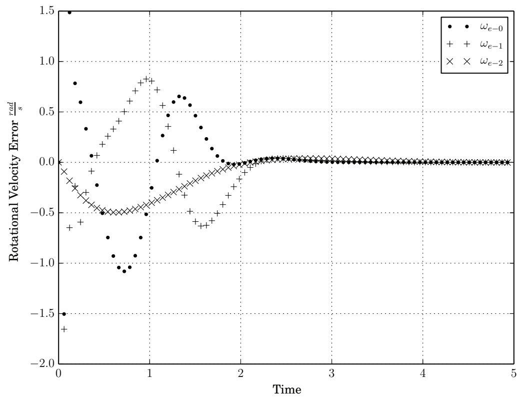

Fig. 9. Quadcopter attitude velocity error (ω_e).

V. Test Bench & Flight Test

Our prototype is a quadcopter assembled at the UMI LAFMIA laboratory. This prototype is composed as follows: the structure is a carbon fiber frame. The four motors are 3525-650Kv 14-Pole Brushless Motors, they are controlled using four 30 Ampere ESC Multi-rotor Motor Speed Controllers. The radio-transmitter has 9 PPM channels working 3.4 GHz. The battery is a 5000mAh 4-cell 14.8V Li-Po Battery. Also, the processor is a 32Bit Autonomous Vehicle Micro-computer, see Figure 10.

Fig. 10. Quadcopter platform.

The processor used is based on the open hardware Pixhawk board, [13], which is an "all in one" unit that combines all the necessary sensors to stabilize the system in attitude, as well as several interfaces for other kinds of inputs and outputs. It has a 168 MHz Cortex M4F CPU with 156 KB RAM, and 2 MB Flash, it includes 3D accelerometers, gyroscopes, magnetometers, and barometers.

A. Flight test

The stabilization of the vehicle is achieved by implementing the dual quaternion based control law (14). The following graphs present the vehicle's position, attitude, orientation trajectory, attitude error, angular velocity and input torques in Figures 11 - 16.

The vehicle was set to stabilize its linear velocity to zero using the barometer to feedback the speed and position in the z axis, and an optical flow sensor for the linear velocity in the x and y axes, no position feedback was tested for the x and y axes because the integration of the optical flow sensor presented important errors, a better sensing technique is needed to make a feedback in position for this axes such as SLAM algorithms, or Opti-Track cameras. The radio controller was used only to elevate the vehicle to the desired altitude and to start the control law.

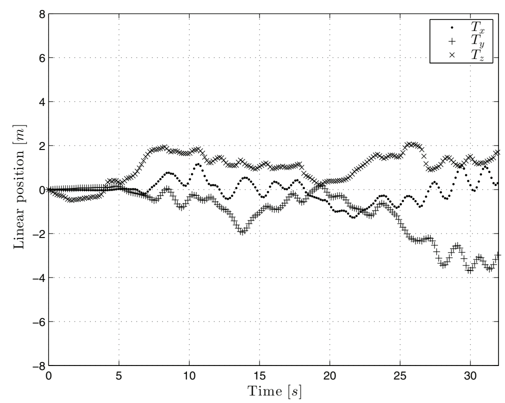

Fig. 11. Quadrotor's position

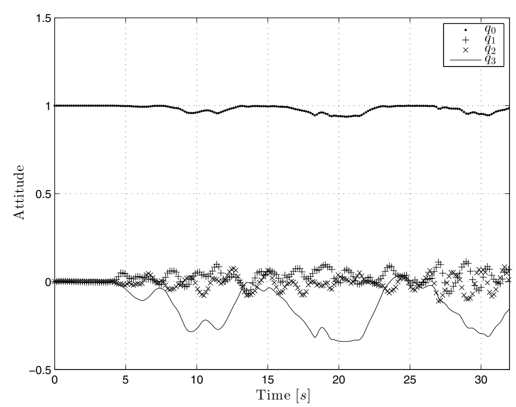

Fig. 12. Quadrotor's orientation

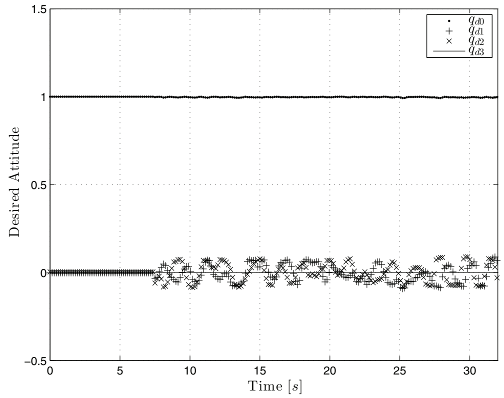

Fig. 13. Quadrotor's orientation trajectory

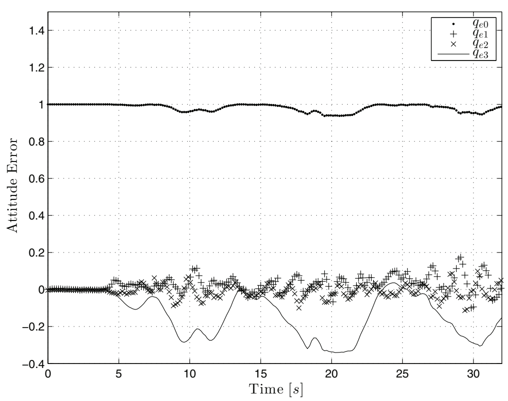

Fig. 14. Quadrotor's orientation error

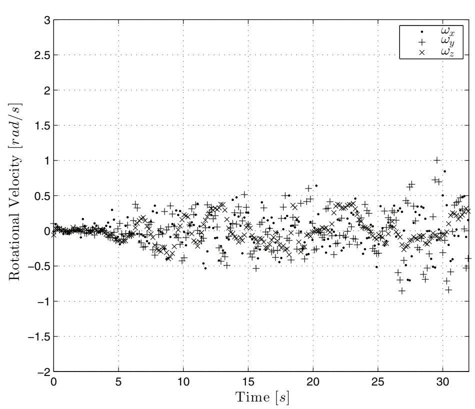

Fig. 15. Quadrotor's rotational velocity

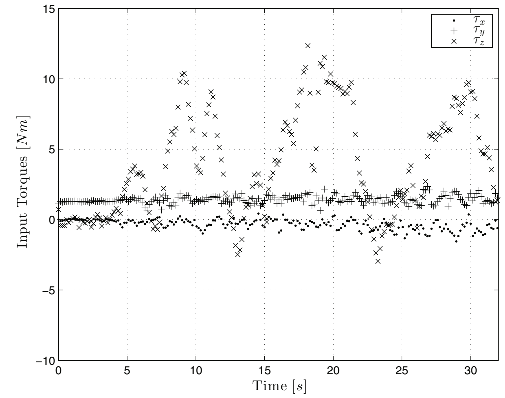

Fig. 16. Quadrotor's input torques

B. Future Work

The stability of the control law was proven for linear velocity in an outdoor environment. A better position estimation technique such as SLAM algorithms, or opti-track systems must be fed to this control law in order to stabilize the position of the vehicle. In this test, a translational reference will be given using the radio transmitter. The speed will be sensed with an optical flow sensor camera, the altitude will be sensed with a barometer, and the $x - y$ position will be estimated with an extended Kalman Filter so that the vehicle stabilizes at the desired position.

Since the rotational and the translational model have an algebraic relationship given by equation (30), it is expected that the attitude system converges to the trajectory needed to stabilize towards the desired velocity.

VI. Conclusion

A dual quaternion model for the quadrotor vehicle is presented in this paper. The dynamic model was simulated in Python programming language, and successfully stabilized using a PD controller. Although only the translational velocity stabilization was implemented in the experimental platform, the results from the simulations and from the flight tests let us foresee promising results for future work.

An exact linearization is also realized in order to apply linear control algorithms. Moreover, it was shown that the quadrotor system can be stabilized asymptotically using simple state feedback controls by taking advantage of the mathematical simpleness of the dual quaternion exponential mapping. The test showed that, even in outdoor environments, the attitude remains stable using the proposed control law.

In addition, the usage of quaternions also eliminates undesired effects on the platform such as the gimball-lock or discontinuities, which are common problems using traditional approaches. This makes possible to accomplish complex tasks and applications such as surveillance using gimbaled cameras or object manipulation.

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